SOKOINE UNIVERSITY OF AGRICULTURE

School of Engineering and Technology / College of Natural and Applied Sciences

Department of Informatics and Information Technology (DIIT)

BIT 111: MICROCOMPUTERS AND APPLICATIONS

TAKEHOME HOMEWORK ASSIGNMENT 1 (TOPIC 1)

Theme: INTRODUCTION TO COMPUTERS, GENERATIONS, PIONEERS & CLASSIFICATION

Course Instructor: Madam Bertha Msuliche Lebalwa (MSc)
Curriculum Coverage: Subtopic 1.1 (Computer Generations), Subtopic 1.2 (History & Pioneers), Subtopic 1.3 (Classifications)
Assessment Weight: Coursework Component (Official Scheme: Assignments 5%, Tests 15%, Practical Report 10%, Practical Test 30%, UE 40% - subject to instructor review)
Submission Deadline: End of Week 2 by 17:00 EAT | Total Allocation: 100 Marks

INSTRUCTIONS TO CANDIDATES:
1. This assignment must be solved individually. Plagiarism, peer copying, or unreferenced AI generation will result in an automatic score of zero and disciplinary reporting under SUA Examination Regulations.
2. Answer ALL questions in Sections A, B, and C. Allocate your effort according to the mark allocations shown in brackets.
3. Support your explanations with clear examples, labeled diagrams, or command syntax where requested.
4. Use spreadsheet formulas only where requested in the spreadsheet assignment; no other numerical calculations are required.
5. Submit your completed work as a clean, stapled technical report or upload the single consolidated PDF to the SUA Virtual Learning Environment (Moodle) before the deadline.

SECTION A: FOUNDATIONAL & HISTORICAL CONCEPTS [25 MARKS]

Answer all five questions in this section. Concise, technically accurate answers are required.

Question 1 [5 Marks]: Differentiate clearly between data and information within an agricultural enterprise context. Illustrate your answer by outlining the four stages of the Information Processing Cycle (IPOS) as applied to an automated grain moisture monitoring station at SUA Mazimbu Farm.

Question 2 [5 Marks]: Compare First-Generation and Second-Generation computers across the following specific dimensions: (a) Primary active switching electronics, (b) Primary internal memory technology, (c) Typical physical power and cooling requirements, and (d) Primary programming paradigm. Provide one canonical machine example for each generation.

Question 3 [5 Marks]: Analyze the historical contribution of Charles Babbage and Augusta Ada Lovelace. Why is the Analytical Engine universally recognized as the theoretical prototype of modern general-purpose digital computers, and what conceptual breakthrough did Lovelace contribute regarding algorithmic execution?

Question 4 [5 Marks]: Explain the fundamental architecture proposed by John von Neumann in his 1945 EDVAC report. What is the 'stored-program concept', and what physical phenomenon constitutes the 'Von Neumann Bottleneck' in contemporary microcomputer processing?

Question 5 [5 Marks]: Distinguish between Analog, Digital, and Hybrid computing systems based on how they represent and process information. Provide an example of a hybrid computer in agricultural engineering.

SECTION B: COMPUTER CONCEPTS & APPLICATIONS [35 MARKS]

Question 6 [10 Marks]: Describe how a processor, memory, storage, and operating-system software work together when a student opens and edits a document. [10 Marks]

Question 7 [10 Marks]: Compare general-purpose, special-purpose, and embedded computers. Give an appropriate example from agriculture, animal science, tourism, engineering, or food science for each type. [10 Marks]

Question 8 [8 Marks]: Explain the stored-program concept and describe one way it changed the design and use of computers. [8 Marks]

Question 9 [7 Marks]: Choose a suitable computer type for an agricultural extension team working both in an office and in the field. Explain your choice using portability, connectivity, software needs, and data protection. [7 Marks]

SECTION C: APPLIED ENTERPRISE & INFRASTRUCTURE DESIGN [40 MARKS]

Question 10 [15 Marks]: The SUA School of Engineering and Technology intends to procure new computer infrastructure for two distinct facilities: (Facility 1) A Computer-Aided Design (CAD) and drone aerial photogrammetry laboratory for precision farming; (Facility 2) An automated field weather telemetry network spanning 12 outlying experimental agricultural stations in Morogoro Rural.
(a) Specify and justify the optimal computer classification (e.g., Workstation vs Desktop PC vs Single-Board Computer vs Embedded Microcontroller) for each facility. [5 Marks]
(b) Produce a comparative specification matrix evaluating both choices across: Processor class, memory capacity and ECC requirements, graphics accelerator, thermal/environmental tolerance, power source, and operating system. [6 Marks]
(c) Outline the network role (Client, Peer, or Dedicated Server) each machine plays in the overall enterprise data collection pipeline. [4 Marks]

Question 11 [15 Marks]: Contrast the architectural paradigms of Supercomputers, Enterprise Mainframes, and Microcomputer Clusters.
(a) Discuss why a commercial banking transaction processing system (such as CRDB or NMB in Tanzania) relies on IBM z16 Mainframes with 99.999% ('five nines') availability rather than

high-performance supercomputers. [6 Marks]
(b) Contrast the concepts of Reliability, Availability, and Serviceability (RAS) with Peak Floating-Point Operations Per Second (FLOPS). [5 Marks]
(c) Explain how a cluster of commodity microcomputers (Beowulf cluster) running Linux can economically perform agricultural genomics and climate forecasting previously restricted to monolithic supercomputers. [4 Marks]

Question 12 [10 Marks]: Terminal Diagnostics & Operating Environment Analysis:
A systems technician at SUA is auditing a microcomputer laboratory running Ubuntu Linux.
(a) Provide the exact terminal commands required to display: (i) Detailed CPU architecture and cache topology, (ii) Total and available RAM in human-readable megabytes/gigabytes, (iii) Motherboard BIOS/firmware vendor and version, and (iv) Storage block device hierarchy. [5 Marks]
(b) Explain what the processor model, core, and thread entries in `lscpu` describe, and why an operating system may report more logical processors than physical processor packages. [5 Marks]